

I. INTRODUCTION:

A "SPRITE" is a group of pixels in the form of a rectangular grid (8x8 low res; 16x16 high res) which you can control as a single unit. By specifying which pixels to turn on or off within your grid you can create graphic shapes. These shapes can be easily colored and animated on your graphics screen.

Spectra Sprite Editor is a powerful programming utility that takes the drudgery out of graphics programming. With this utility you will be able to easily take advantages of the powerful sprite capabilities of your Spectravideo computer.

The applications of Spectra Sprite Editor will be appreciated if you have some understanding of BASIC programming. However, even if you are just starting to explore programming in BASIC, you will find Spectra Sprite Editor fun and exciting to work with. More importantly, it will stimulate your motivation to learn more and understand advanced programming concepts. This manual does not intend to

serve as a BASIC tutorial.

We assume that terms such as "SPRITE", "BYTE", "BIT", "RAM" are somewhat familiar to you and that you have already experienced working with your SPECTRAVIDEO computer. The few BINARY and DECIMAL notations included in this manual are those regarding directly to the creation of SPRITES.

A. OVERVIEW

This program will enable you to create sprites in both high (16x16) or low (8x8) resolution on your monitor or TV screen, using the built-in joystick on your SV-318 (or cursor control keys on your SV-328). Your sprites can then be saved to cassette as data files to be incorporated into your own program. If you have ever created sprites using long data statements you will appreciate the convenience of actually just drawing your sprite and have the computer program the data values for you.

B. EQUIPMENT NEEDED

All you need to work with Spectra Sprite Editor is:

1. Color television or color computer monitor
2. SV-318 or SV-328 computer
3. SV-902 data cassette recorder
4. Blank cassette tape (for saving your data files)

C. A WORD ABOUT CASSETTES

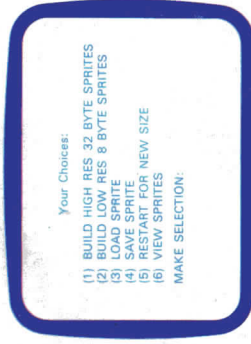
Make sure your program cassette containing "Spectra Sprite Editor" is fully rewound before attempting to CLOAD the program. After you create your own sprites you will probably need to store them on cassette for incorporating into your own program. For this we recommend using a separate blank cassette so as not to accidentally erase your original program cassette.

D. LOADING SPECTRA SPRITE EDITOR

Insert your program cassette (make sure it is fully rewound) then type CLOAD "SPRITE" or just CLOAD with no file name (by just typing CLOAD the first program will be loaded by default), then press <ENTER>. Very quickly the computer will print out "FOUND: SPRITE". It will then take a minute or so for the program to

be loaded. When the "OK" prompt appears on the screen type RUN and press <ENTER> or simply press function key 5 (F5).

You are on your way! You will then see the title page "SPECTRA SPRITE EDITOR" and the classic sprite demo jets flashing across the screen. The program will then display "YOUR CHOICES", which is your main menu for working with "SPECTRA SPRITE EDITOR".



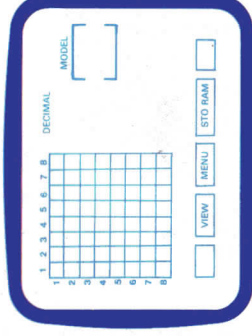
II. CREATING SPRITES:

A. CREATING 8 BYTE SPRITES

To create your first sprite begin by selecting choice 2 from the main menu (press 2). When you choose the 8 byte sprite option the computer will bring up the 8x8 sprite editor. To the right of the 8 by 8 block grid you will see "decimal", and "model" with a small frame below. To create your sprite, move the cursor using the built-in joystick (or cursor control keys on the SV-328) to the box you wish to turn on in your 8x8 grid. Once the cursor is positioned over the box you have chosen, press the SPACEBAR to turn it on. If you wish to turn it off press the SPACEBAR again. Notice that if you keep pressing the SPACEBAR, the box under the cursor will be turned on and off alternately. Continue to turn on other boxes, thus completing your sprite. A scale model of your sprite is being displayed in the "model frame" to the right of your work area as you create your sprite.

When the sprite is just the way you want it,

press function key 4 (F4) to store your sprite data in memory. At this time, you must specify the area in RAM where you wish it stored. If this is your first sprite, store it in location 0.



The following area is for the use of advanced programmers. It contains material that may not be easily understood by beginning BASIC programmers, but can easily be used by persons familiar with binary and decimal notation. After

its value --> 128 64 32 16 8 4 2 1
1 byte --> 1 1 1 1 1 1 1 1
Bit number --> 7 6 5 4 3 2 1 0

If a byte has the bits shown below equal to 1
not 0, it's value is equal to 128.

BIT VAL 128+ 64+ 32+ 16+ 8+ 4+ 2+ 1

BYTE 1 0 0 0 0 0 0 0 BYTE VAL

BYTE 2 0 0 0 0 0 0 0 = 128 (128+1)

BYTE 3 0 0 0 0 0 0 0

BYTE 4 0 0 0 0 0 0 0

BYTE 5 0 0 0 0 0 0 0

BYTE 6 0 0 0 0 0 0 0

BYTE 7 0 0 0 0 0 0 0

BIT # 7 6 5 4 3 2 1 0

Notice the 8 by 8 grid. This represents 8 bits

across (the 8 bits across make 1 byte) by 8 down. The bits that are turned on are equal to the bit values listed above the grid. The total of the 8 bits (don't forget 8 bits make 1 byte) are equal to the decimal equivalent to right of their byte. The bits that are not turned on have a value of 0.

At this time you can either go to the main menu or view the sprites. Let's view your sprite, press function key 2 (F2) and in a few seconds you will be in the view mode. Now you can move your sprite around the screen with the built-in joystick for cursor control keys on the SV 328). You can do much more in the View mode, but we will cover that later. Let's return to the main menu by pressing function key 3 (F3).

B. CREATING A 32 BYTE SPRITE

If you have just finished creating an 8x8 byte sprite, press selection #5 from the main menu to reinitialize the computer.

NOTE you cannot mix high res (32 byte) and low res (8 byte) sprites.

Now choose selection #1 to build a high res

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Tips on Handling and Storage

1. Avoid touching the tape with your fingers; you may cause drop-outs or scratches.
2. Protect your cassette from dust and damage by not in use by keeping it in its protective plastic box.
3. The tape should be wound snugly on the cassette reels; if the tape is loose, carefully wind it on to one of the reels by gently turning one of the hubs.

4. Keep the recorder head and pinch roller clean by regular use of a good cassette head cleaner.

5. Avoid prolonged exposure of cassettes to excessive heat, sunlight and strong magnetic fields. Do not store near heating appliances, TV sets or similar equipment.
6. To preserve valuable or prized recordings, punch holes and remove the erasure prevention lug on the top edge of the cassette.

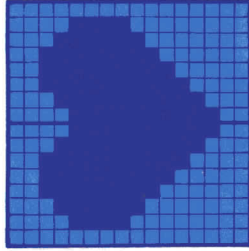
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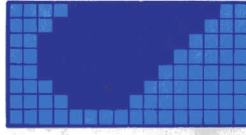
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or 32 byte sprite (16x16). A 32 byte sprite and an 8 byte sprite are both constructed the same way. The only difference between the two is that the 32 byte sprite offers 256 individual points that can be turned on or off as opposed to the 64 points in the 8 byte sprite. Make a sprite in the high res mode and store it in RAM by pressing function key 4. Store it in location 0. Make another sprite and store it in location 1. Now make a third sprite, store it in location 2.



Due to the fact that one BYTE is only 8 BITS long, each row of 16x16 BIT sprite consists of 2 BYTES. The way it is stored in memory is that the 16x16 sprite is split into two columns. The left column is stored in memory first after which the right column is stored in memory. (See illustration)



1)



2)

NOTE on storing sprites - When you press function key 4 to store your sprite you will see "Highest area used=". If you are storing your first sprite, "Highest area used" will equal 0 because no sprites have been stored. When you are saving your second sprite, "Highest area used" will equal 0 because your first sprite is stored in location 0. When you are prompted for the location to store your second sprite, choose location 1 so as not to delete the sprite in location 0 etc.

Now enter the View mode by pressing function key 2 (F2).

III. VIEWING SPRITES

After creating your sprites and storing them in memory, press function key 5 (F5) to view them. You can also view sprites that have been loaded into RAM from a data cassette (as long as they were created by Spectravideo Sprite Editor).

A. **FORWARD (F1)** - If you wish to view the sprites you have created, you may

individually view them in ascending order by pressing (F1).

B. **BACK (F2)** - Allow you to view sprites in descending order.

C. **MENU (F3)** - Return you to the Main menu.

D. **COLOR (F4)** - Let you change the color of your sprite to any of the 15 possible colors. Pressing F4 will cycle you through the different colors.

E. **SIZE (F5)** - This key will toggle you between the enlarged or reduced sprite size. This works in either the 8x8 or 16x16 sprite mode.

F. **JOYSTICK** - The built-in joystick on the SV-318 or the cursor control keys on the SV-328 will enable you to move your sprite on the screen.

G. **BACKGROUND COLOR (ESC)** - By pressing the escape key you can cycle through colors to change your background color.

IV. SAVING AND RETRIEVING SPRITES:

A. **SAVING SPRITES** on cassette - To save your sprites to cassette: return to the main menu, then:

1. Place your cassette into your data cassette recorder. If your cassette has a leader (or transparent tape) at the beginning, make sure that you are past the leader before processing to the next step.

2. Choose selection #4 from the main menu (by pressing 4).

3. Answer the screen prompts for file name and sprite numbers to be saved.

B. **RETRIEVING SPRITES** from cassette - To retrieve sprite data: choose selection #3 from the Main menu. Make sure your data cassette tape is positioned slightly before the beginning of the file you wish to load.

V. USING SPRITES IN A PROGRAM:

In order to use the sprites you have saved, it is necessary to incorporate a short file-loading routine into your own program. A sample program

that loads a sprite file and displays them on the screen is listed following with a short explanation.

```

DEMO 1 "Sprite Loading Sub-Routine"
10 PRINT "NAME OF FILE TO LOAD": INPUT
   N$
20 OPEN "CAS:"+N$ FOR INPUT AS 1
30 INPUT #1, RES
40 INPUT #1, NUM
50 IF RES=8 THEN R1=0 ELSE R1=2
60 SIZE = 1: T = 0
70 SCREEN 1, R1+SIZE
80 FOX X = 1 TO NUM
90 SS=""
100 FOR Y = 1 TO RES
110 INPUT #1, A
120 SS=SS+CHR$(A)
130 NEXT Y
140 SPRITES(X) = SS$
150 NEXT X
160 FOR X = 1 TO NUM
170 PUT SPRITE 0, (T, 95), 9, X
180 T = T+1

```

```

190 FOR Y = 1 TO 50:NEXT Y: NEXT X
200 GOTO 160
210 END

```

Remarks about above program: Numbers indicate line numbers in program.

20 The "OPEN" statement opens a data file. "CAS" specifies that it's on the cassette recorder.

30 RES is the resolution size that the sprites in the file are stored in. This along with NUM (line 40), were written to the file by the Sprite Generator Program when the file was saved.

40 NUM is the number of sprites in the file.

60 SIZE is the size of the sprite. This can be changed by the program to switch between large and small sprites.

70 Screen 1 means graphics mode. (R1+SIZE) specifies large or small size sprites. Use 1 & 2 for 8x8 (8 byte) sprites and 2 & 3 for 16x16 (32 byte) sprites.

80 X will increase for each sprite on the file.

100 For each byte of the sprite location.

110 We will read one byte.

120 "Add" the bytes together.

130 And go back and get another byte.

140 Now that we have the whole sprite we will

identify it by the current value of X.

150 Now we do the same procedure for the next

sprite.

160 This NEXT section will put the sprites at a

specific location on the screen. For a

different location, simply change the

coordinates in the parentheses.

180 This particular program will put all the

sprites on the screen (NUM).

190 This program will put the sprites at location

(T, 95) on the screen. ...Read your user's

guide!

A. Type the sample program (demo 1) into your

computer.

B. Place the cassette with the sprite file you

have created using Spectra Sprite Editor

into the data cassette player and properly

position the tape.

C. Type "RUN" and follow the program

directions that will appear on the screen.